Selected Issues in History and Philosophy of Science 3240305

Spring 2026

Lecturer: Dr. Alik Pelman, alikpelman@technion.ac.il

Times and venues: Tuesdays, 15:30, Ullmann room 604

Exams (tentative!): Term A: July 9th; Term B: August 11th.

Credits: 2 Prerequisites: none

Course Description and Objectives

In 1987, the US Supreme Court rejected the requirement to teach creationism in schools, ruling that, unlike evolution theory, creationism does not meet scientific criteria. Could you participate in such a debate and defend your position? What makes a theory scientific, and what distinguishes it from non-scientific theories? Can you critically evaluate justified critiques of science and defend science against unjustified criticisms?

This course focuses on the scientific method and criticisms against it. The first part explores the scientific ideal through three main aspects: (a) the ideal method of developing scientific theories and using them to explain and predict natural phenomena; (b) the ideal relationships between theories across various sciences (e.g., biology and physics); and (c) the ideal relationship between new and past theories within the same field. We will emphasize the central role of logic in ideal scientific practice. The second part of the course examines significant criticisms of this scientific ideal method, many of which remain unanswered. We will primarily explore critiques "from within" science, identified by members of the scientific community themselves, and then briefly address "external" criticisms from non-scientific perspectives (such as cultural criticism and religion).

Learning Outcomes

By the end of the course, students will have a thorough understanding of the scientific method. They will be able to apply scientific logic to: formulating theories from observations; verifying and falsifying existing theories; and understanding explanatory relationships between theories and phenomena. Students will develop a nuanced perspective on the strengths and limitations of science, gaining tools for impartial reflection on their research fields. Additionally, students will be equipped to engage with critiques of science, including those from alternative and postmodern viewpoints, distinguishing clearly between scientific and non-scientific explanations.

Course Schedule

Meet	Subject	Content	Reading
1.	Introduction	Introduction and administration. Course syllabus. **Introduction**: The ideal scientific method; Logical Positivism	
2.	The Scientific Method (Positivism)	**The Scientific Method**: The relation beteween theory and observation; The relations between theories.	Hempel, Ch. 2
3.		**Logic of Science - Introduction:** Deductive, inductive, and abductive arguments.	Priest, Videos
4.		**Logic of Science I: From observation to theory;** generation; falsification; confirmation. Popper's falsification principle.	Popper, 'Science: Conjectures and Refutations'
5.		**Logic of science II**: B. From theory to observation: explanation, prediction. C. Relations between theories: synchronic; diachronic.	Hempel, Ch. 5
6.	Critique of Science (internal)	**Criticism of the method of confirmation**: Problems of induction; Ravens Paradox; The New Riddle of Induction.	Godfrey-Smith, Ch. 3
7.		**Criticism of the method of falsification**: Duhem–Quine thesis. Challenges to Popper's insights. Criticism of the to the method of generation: a. by induction; b. by abduction.	Duhem, *Aim and Structure*
8.		**Criticism of the method of Prediction**: Challenges to laws of nature.	Chalmers, A., Ch. 14
9.		**Criticism of the method of explanation**: Challenges to the deductive-nomological model.	Salmon, 46-50
10.		**Criticism of the Unified Science Ideal**: Criticism of the reductionist approach.	Heil, 69-80
11.		**Kuhn's Alternative**: Kuhn's criticism of the method of theory replacement. **Summary**	Kuhn, Ch. 1

Assessment Methods – Final Grade Composition

Final Exam (multiple-choice): 90%

Very short weekly exercises: 10%

Course Requirements and Policies

Reading: Each lesson will include a brief question about the next lesson's reading. Each submission grants one point toward the final grade, totaling 10 points.

Textbooks

- Chalmers, Allen, *What Is This Thing Called Science*
- Godfrey-Smith, Peter. *Theory and reality: an introduction to the philosophy of science*
- Okasha, Samir. Philosophy of Science

Primary Sources

- Duhem, Pierre, *The Aim and Structure of Physical Theory*
- Hempel, Carl. *Philosophy of Natural Science*
- Kuhn, Thomas. *The Structure of Scientific Revolutions*.
- Popper, Karl. "Science: Conjectures and Refutations"
- Salmon, Wesley. Four *Decades of Scientific Explanation*
- Heil, John. *Philosophy of Mind*